



Blood Bank Management System

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Abstract

- Blood Bank Management System manages blood donor details, blood groups, and available blood units efficiently.
- It helps quickly search, store, and update blood stock information during blood requests and emergencies.

Problem Statement & Objectives

- Problem:** Manual blood stock management can cause delays, errors, and difficulty in finding available blood.
- Objective 1:** Manage donor and blood group information efficiently.
- Objective 2:** Track available blood units using DSA concepts.
- Objective 3:** Quickly search and process blood requests.
- Scope:** Donors, blood inventory, blood requests, blood search, and admin management.

Existing vs Proposed System

Feature	Existing System	Proposed System
Donor Management	Manual records	Digital records
Blood Search	Time-consuming	Fast searching
Stock Management	Manual updates	Automatic updates
Blood Requests	Manual processing	Organized request handling
Limitation / Gap	Slow and error-prone	Faster and efficient using DSA

Feasibility Study

- Technical:** React, Node.js/Express and MySQL provide a suitable web stack for the Blood Bank Management System.
 - Economic:** Open-source technologies and free development tools reduce the overall project cost.
 - Operational:** Simple web interface for donors, admins and hospitals; replaces phone calls and paper records.
- Schedule:** Modules can be developed and tested incrementally within the semester timeline

Results & Testing

Successfully stores and manages donor details.
Quickly searches blood groups and availability.
Performs CRUD operations correctly.
Updates blood requests and manages stock.

Conclusion

- The system provides an efficient way to manage blood donors, stock, and requests.
- DSA concepts help improve searching, storing, and processing blood information.

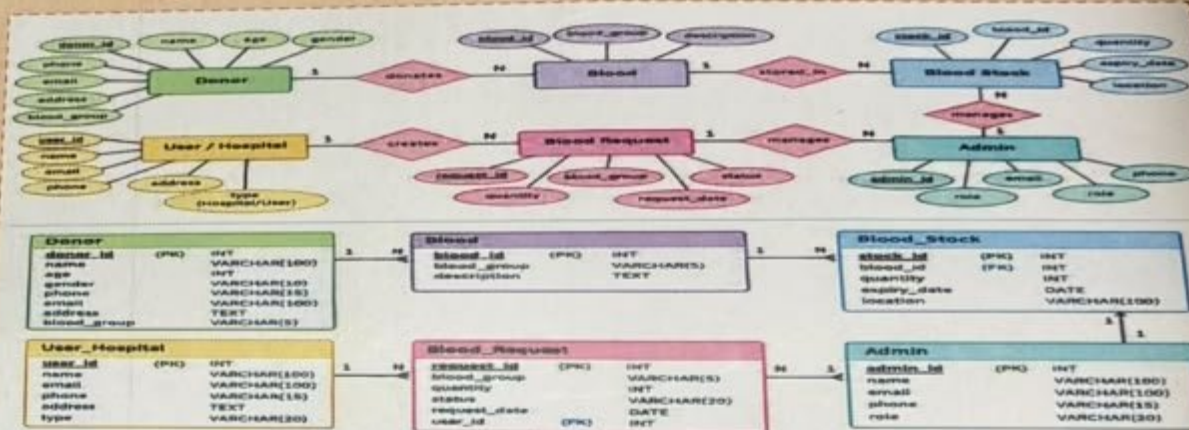
Tools & Technologies

- Front-End: React.js
- Back-End: Node.js + Express.js (REST APIs)
- Database: MySQL
- Tools: Git & GitHub, Postman, VS Code

System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Add online blood request and emergency alerts.
- Add hospital and donor location tracking.
- Develop a mobile application.